
3AL Test 1 2023

Time: 90 minutes

Available marks: 70

Full marks: 65

Instructions

1. Answer all questions in the lined answer booklets.
2. You can answer the questions in any order but start each question on a new page.
3. Make sure each answer is clearly labelled.
4. Show all work and reasoning unless otherwise instructed.
5. This test consists of **4** questions. Make sure your question paper is complete.
6. If you have read the instructions, draw a star on the front of your answer booklet.
You will receive an extra mark for doing so.

Good luck! $\diamond_+^+$

1. Let G_1 and G_2 be groups. Use the First Isomorphism Theorem to prove that there is a factor group of $G_1 \times G_2$ that is isomorphic to G_1 . Clearly show what the factor group is.

NOTE: This question is asking you to find a normal subgroup $H \triangleleft G_1 \times G_2$ such that $(G_1 \times G_2)/H \cong G_1$.

[15 marks]

2. (a) State the Second Isomorphism Theorem.
(b) Let G be a group of order $p^n q$, where p and q are distinct primes and $n \geq 1$. Let K be a normal subgroup of G of order p^n . Suppose H is a subgroup of G .
 - (i) Show that $|HK/K|$ divides q .
 - (ii) Show that if $|H| = p^n$, then $|HK/K| = 1$ and $|H \cap K| = p^n$.
 - (iii) Explain why K is the only subgroup of G with order p^n .

[5 + 5 + 10 + 5 = 25 marks]

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3. Let G be a group with normal subgroups H and K such that $H \subseteq K$. Prove or disprove: If G/H is cyclic, then so are G/K and K/H .

[10 marks]

4. Consider the group $G = \mathbb{Z}_{250}$ with subgroup $K = \langle [10] \rangle$. Note that $250 = 2 \cdot 5^3$.

- (a) Determine all subgroups of $\mathbb{Z}_{250}$ and draw the lattice diagram.
- (b) Use the Correspondence Theorem to draw the lattice diagram for $\mathbb{Z}_{250}/K$.
- (c) Let $H = \langle [50] \rangle$. Determine the number of subgroups of $(\mathbb{Z}_{250}/H)/(K/H)$.

NOTE: You can use the fact that isomorphic groups have the same number of subgroups (no need to prove this).

[10 + 5 + 5 = 20 marks]